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Automation of the Dewatering Operation for Crude Oil Storage Tanks

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Abstract

This paper describes the implementation of an Automatic Dewatering System in one of ADNOC Offshore's crude oil storage tanks, carried out as a pilot project. The system automates tank dewatering, reducing Health, Safety & Environmental (HSE) hazards and removing the need for standby operators during the process.

Automated dewatering in crude oil storage tanks is achieved by installing two oil-water interface detector probes in each draw-off sump, resulting in a total of eight probes distributed across four sumps. These probes operate in conjunction with both the Local Control Panel and the Distributed Control System (DCS) to manage the dewatering process automatically. The primary probe, designated as the Control Probe, is situated at the upper section of the draw-off sump and governs the actuation of the water draining motor-operated valve (MOV) based on the detected oil-water interface level. The secondary probe, referred to as the Alarm Probe, is installed at the lower portion of the sump and serves as a safety feature by triggering an alarm and ensuring closure of the MOV should the Control Probe malfunction. This arrangement significantly reduces the risk of oil contamination in the water drain line. As a result, the system reliably discharges only clean, oil-free water from the tank via the dewatering lines connected to the sumps.

Automatic dewatering technology was successfully integrated into an existing Crude Storage Tank, with installation and commissioning completed without interruption to ongoing operations. Comprehensive testing during the Site Acceptance Test (SAT) encompassed multiple operational scenarios, all of which yielded satisfactory results and confirmed effective system integration. The technology substantially reduced oil carryover, consistently achieving the process guarantee of 50 ppm oil in produced water.

The system has operated efficiently for over nine months, reducing the need for full-time field operators and lowering operating costs and HSE risks during dewatering. Its successful implementation in one of ADNOC Offshore's crude oil storage tanks automated the dewatering process, eliminated standby operator requirements, and paved the way for adopting similar technologies in other tanks.

Introduction

Storage tanks containing crude oil and refined products accumulate water at the bottom as a result of hydrocarbon and water separation within the tank. During crude tank operations and loading procedures, tanks are initially filled with crude oil up to the designated normal or high liquid level, after which they are

allowed to stand for approximately 6 to 8 hours to enable water to settle. Industry standards require periodic drainage of this water. Manual water draw-off frequently results in minor oil carryover at both the start and end of the process, since operators must observe the transition from water to oil in order to close the valve; additionally, residual oil in the draw-off pipe is typically flushed out when effluent water discharge commences. The water removed from the tanks is subsequently routed to the Sour Water De-Oiling Plant (SWDP) for oil recovery.

Draining water from the tank promptly upon reaching the desired quality is essential to prevent the following issues:

- Corrosion of the tank bottom floor
- Boil overs in the event of a fire
- Limitation of storage capacity for hydrocarbons

Timely removal of water ensures the full utilization of the tank's capacity for hydrocarbon storage and enhances operational safety.

The setpoint for water quality, specifically the concentration of oil in water, shall be determined in accordance with local EPA (Environment Protection Authority) regulations and the standard operating procedures governing effluent discharge to drains or treatment facilities. In many jurisdictions, the permissible level of flammable liquid hydrocarbons in effluent must not exceed 50 milligrams per liter (50 PPM), measured as free oil exclusive of dissolved hydrocarbons. Furthermore, dewatering effluent must not create an explosive atmosphere in downstream facilities. It is required to provide a performance guarantee that ensures oil content in produced or settled water, processed through the dewatering system and discharged from the tank to the SWDP (Sour Water De-Oiling Plant), does not exceed 50 ppm.

Manual Dewatering

Manual dewatering from each tank is performed by operators opening the valves on the dewatering lines. The process typically requires 4 to 6 hours, depending on both the water content in the treated crude and the tank's elevation. During this operation, oil carry-over is closely monitored through sight glasses; dewatering ceases immediately upon detection of oil carry-over.

There are several operational concerns associated with manual dewatering:

- Water drainage relies entirely on manual operation and visual observation, making it prone to human error.
- There is an increased risk of oil carry-over into the drained water, which can place additional strain on the Sour Water De-Oiling Plant (SWDP) and Oil Recovery systems.
- Personnel must be engaged throughout the dewatering period.
- The overall dewatering and tank preparation process is prolonged, as settling can only begin after tank filling is complete, followed by manual dewatering.

Automatic Dewatering

A study was conducted and various technologies were compared to automate the dewatering process. The study concluded that installing oil-water interface detector probes in the storage tanks is appropriate for enabling fully automatic dewatering operations without the need for manual intervention.

Tank Nozzle Arrangements

For the installation of probes, each storage tank is equipped with dedicated nozzles—specifically, two nozzles per dewatering sump. In Pilot Storage Tank, a total of eight nozzles (corresponding to four sumps, with each sump having two 2-inch nozzles) were installed during the tank replacement.

Normally each tank are provided with two to four dewatering nozzles depending on the tank size and water content in the stored product, in the pilot storage tank there are 4 dewatering nozzles are provided with piping routed internally to direct flow into the respective sumps. Please refer to [Figure-1](#) for a representation of the dewatering probe and sump arrangement on the tank.



Figure 1—Dewatering Nozzle and Prob arrangement inside the Tank

Dewatering Prob Arrangement

A dewatering system employing absorption-type probes is recognized as one of the most effective methods for dewatering hydrocarbon tanks at oil storage and terminal facilities. This system measures water quality using energy absorption probes to determine the oil (hydrocarbon) concentration within the water designated for removal. The oil-in-water concentration serves as the set point for initiating the dewatering process.

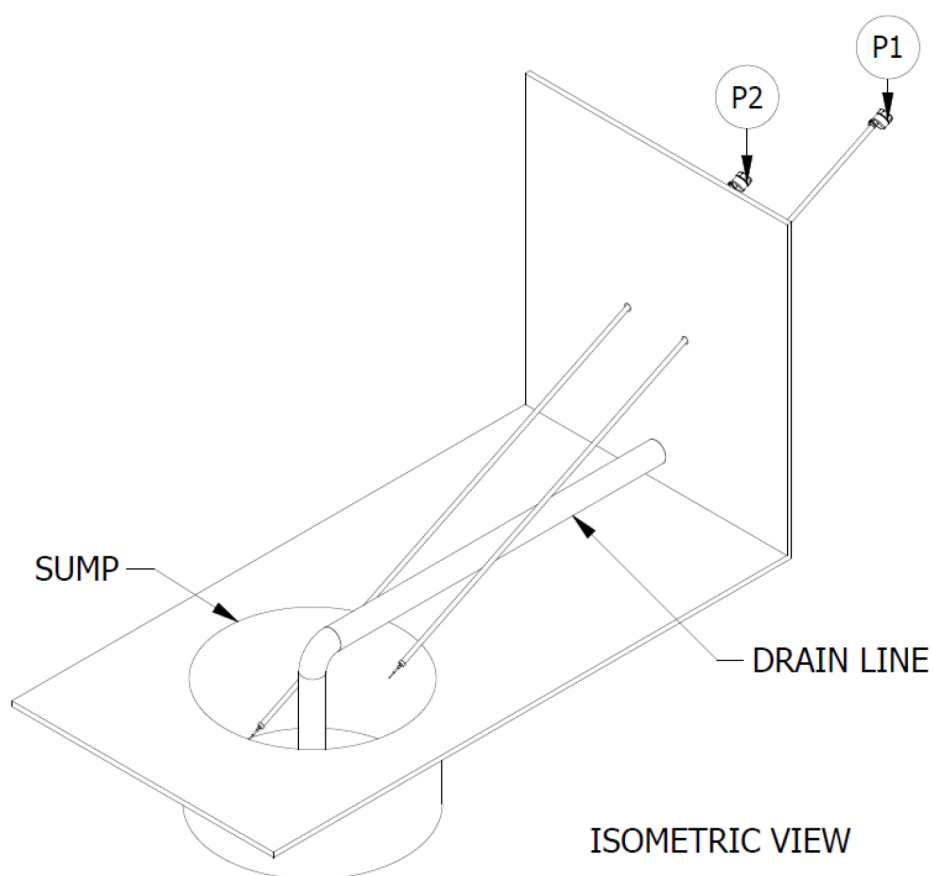


Figure 2—Isometric View of Automatic Dewatering Nozzle and Prob Arrangement

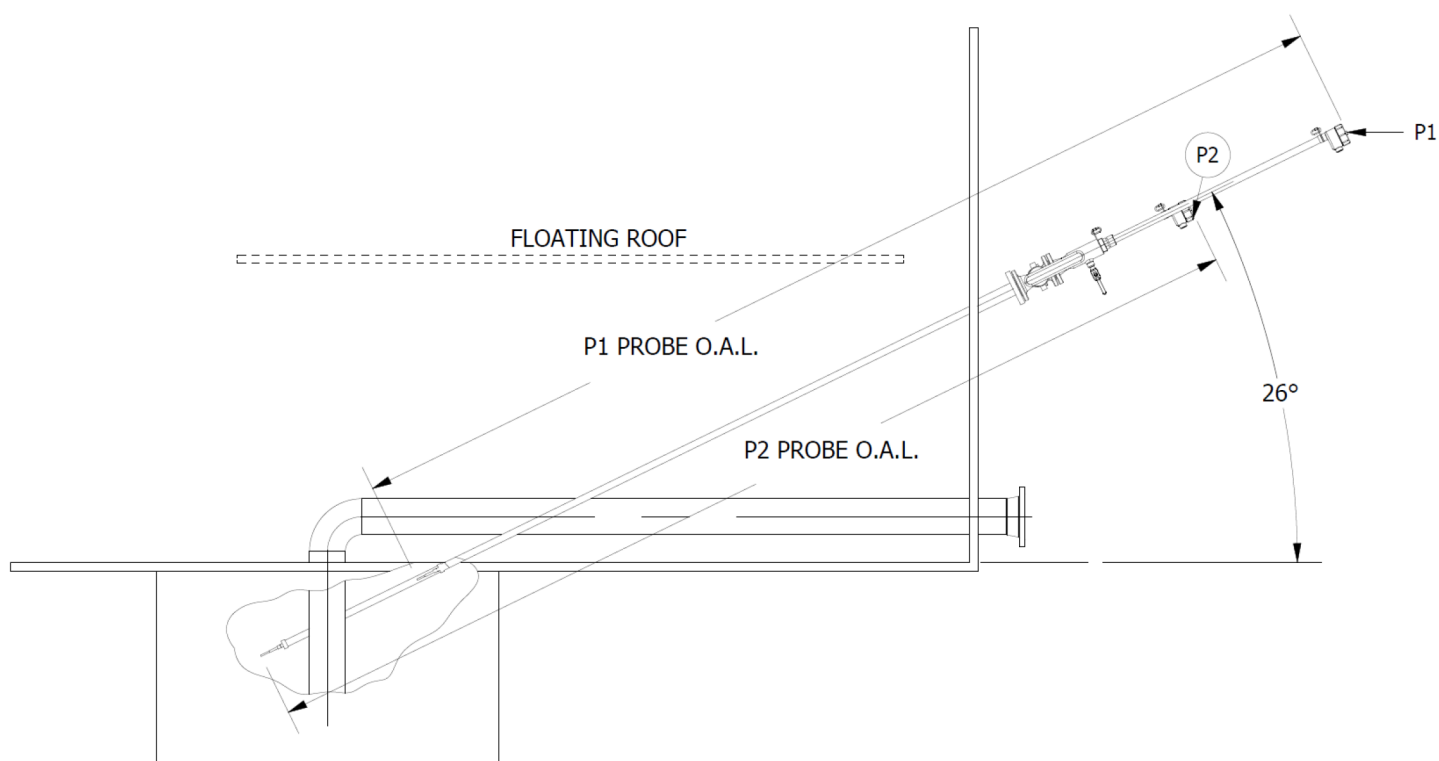


Figure 3—Side View of Automatic Dewatering Nozzle and Prob Arrangement

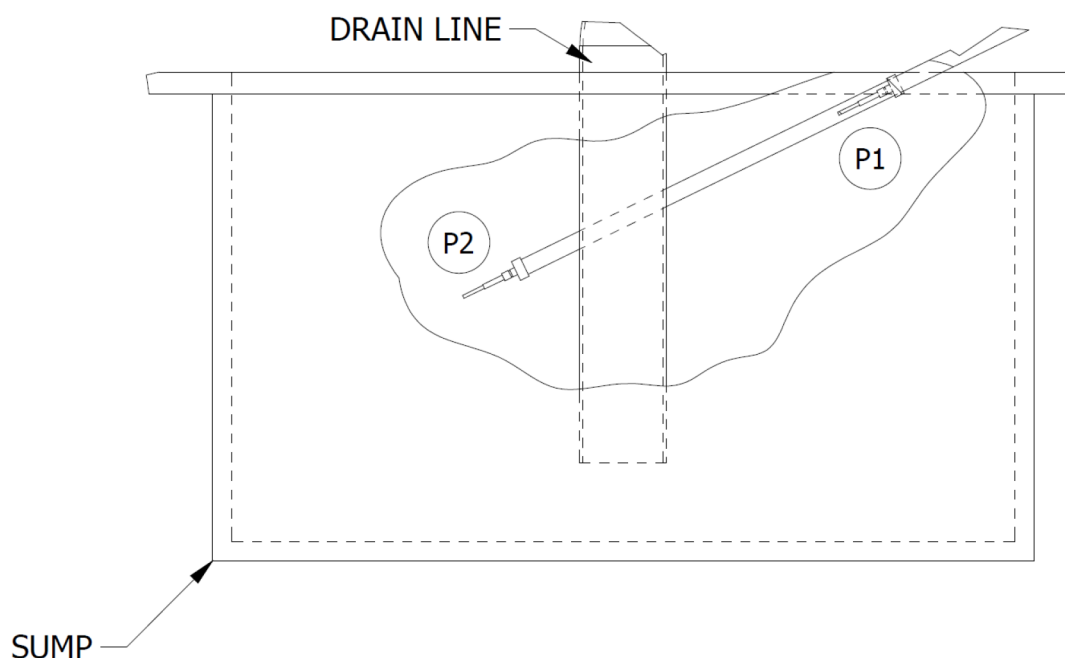


Figure 4—Detail View of Dewatering Probe Arrangements inside the Draw-off Sump

The standard configuration sets the control probe threshold at 80%. When the water concentration in the emulsion drops below this level, the Control Probe (P1) relay closes the motor-operated valve (MOV) through the Distributed Control System (DCS), halting the dewatering operation. The Alarm Probe (P2) is configured with a set point of 90% water cut. If the water concentration falls below this threshold, the alarm probe overrides the control probe, issuing a signal to close the valve and generating an oil alarm to the DCS Human Machine Interface (HMI). It is important to note that the Control Probe exclusively governs the MOV, except when overridden by the Alarm Probe upon detection of oil. Both probes do not simultaneously control the valve. In the event of a valve failure resulting in an alarm, manual intervention will be required to close the valve. Final system settings may require field adjustment during commissioning.

Each tank sump must be equipped with a minimum of two probes:

- Probe 1 (P1) – Control Probe
- Probe 2 (P2) – Alarm Probe

Probes should be mounted at a downward angle, allowing for elevation adjustment and optimal control and alarm settings. Adjusting probe elevation affects the settling time of the tank, enabling attainment of the desired water quality. Raising the oil-water interface enhances water quality but may compromise other operational criteria. Integration with the system enables further optimization of the interface level.

Crude oil tanks are typically designed with sumps to collect water separated from crude oil during the settling phase. Appropriate nozzles should be installed on the tank walls adjacent to the sumps to accommodate probe placement and ensure system performance. Additionally, the tank-dewatering draw-off pipe must be inserted into the sump and engineered to prevent vortex formation at the draw-off inlet, as specified by API 650 standards.

Details of Dewatering Prob

The dewatering probe features a body diameter of 12.7 mm. It is equipped with a removable junction box (J-box) at one end and an antenna at the opposite end. The J-box serves as the housing for the transmitter ("hockey puck"), which is connected to the probe via banana plugs and sockets. A compression fitting attaches the J-box securely to the probe body. Standard probes are manufactured in overall lengths of up

to 10 feet, primarily for installations where the probe is suspended vertically through the tops of vessels in non-turbulent processes. For applications requiring greater length, alternative probes with a body diameter of 31.75 mm are available.

Automatic Tank Dewatering Control Logic

The Tank De-watering System is engineered to detect water accumulation in the sump at the base of the tank and to automatically discharge this water by transmitting a signal to the motor-operated valve (MOV) located on each tank's (or sump's) drain line.

The Automatic Tank Dewatering System comprises the following components:

- Probe-1 (Control Prob): Installed within the tank sump for system control.
- Probe-2 (Alarm Prob): Installed within the tank sump for alarm monitoring purposes.

Control logic is implemented via relays in the Control Unit, as defined by the system's operational logic table ([Table-1](#)). This logic is configured in DCS.

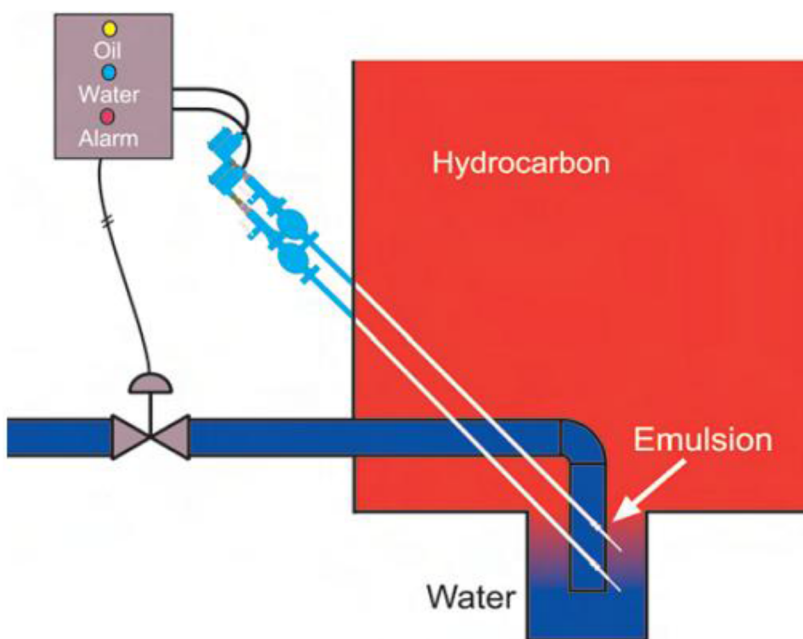


Figure 5—Automatic Dewatering Control Logic

Upon detection of free water by Probe-1, the corresponding relay in the Control Unit is energized, initiating a command to open the water drain/discharge valve. As water is discharged from the tank's bottom, the hydrocarbon/water interface drops. Once Probe-2 detects this interface, its relay de-energizes and signals the closure of the drain/discharge valve.

Under normal operation, the oil-water interface is maintained at the Probe-1 level, ensuring that only water is removed from the sump. The probes measure water concentration by percentage, not in parts per million (PPM), but are designed to maintain outlet water content below 50ppm. Output water quality is highly dependent on the effective separation of oil droplets.

To ensure complete dewatering, both probes must be situated within the tank sump. However, given the sump's limited volume, drainage must occur gradually for two reasons:

- to prevent turbulence and re-mixing of oil and water, and

- to accommodate the delay between water detection by the probe and full closure of the valve.

Rapid drainage risks removing all water prematurely and allowing oil into the drain line. Therefore, the MOV should remain partially open to allow a controlled flow rate.

The system commences dewatering once both probes (control and alarm) detect water, irrespective of any predefined settling period. Field trials have allowed adjustment of settling time and valve opening percentage to achieve optimal draining and guarantee outlet concentrations do not exceed 50ppm.

The dewatering MOV is partially opened to regulate drain speed, minimizing fluid velocity and preventing oil ingress into the drain line. This control depends on both actuator response speed and gravitational fluid velocity. Partial valve opening reduces fluid speed and minimizes closure time. For example, pneumatic valves known for rapid closure can be fully opened, with prompt closure preventing oil contamination. In contrast, MOVs respond more slowly and thus must not be fully opened to avoid oil entering the drain before complete closure. Field testing allows response parameters to be optimized accordingly.

The lower probe (Probe-2 or alarm probe) is designed for continuous operation in clean water and serves as a backup to the main control probe. Should an upset occur or if the water draw-off valve malfunctions, causing oil to reach the sump's base, Probe-2 activates the alarm relay in the Control Unit and alerts operators for corrective action. Under normal conditions, the alarm remains inactive; upon activation, it overrides the control probe, closes the valve, and deactivates the control relay. Alarm activation may indicate MOV failure, requiring operator intervention.

The logic depicted in [Table-1](#) below outlines system operation using two probes as reference.

Table 1—Automatic Tank Dewatering System Control Logic

Probe Reading			Actions			
Case	High Probe (Control)	Low Probe (Alarm)	Water Light	Oil Light	Alarm Light	MOV
1	WATER	WATER	ON	OFF	OFF	OPEN
2	OIL	WATER	OFF	ON	OFF	CLOSE
3	OIL	OIL	OFF	OFF	ON	CLOSE
4	WATER	OIL	OFF	OFF	ON	CLOSE

For automatic tank operations, the reliability of the dewatering control system relies on several factors related to tank condition, control valve performance, and process parameters. Oil and water enter the tank well-mixed, forming dispersed oil droplets beneath the emulsion layer. The rise rate of these droplets and available retention time determine the oil content in discharged water. Larger and less dense droplets ascend faster.

Slower effluent discharge enhances water quality. Thus, optimal tank operation promotes efficient oil-water separation and high-quality water output. Achieving specification-compliant water requires adequate retention time and a slow discharge rate. Since separation is gravity-driven and reliant on retention, any dissolved hydrocarbons unseparated at maximum retention will be discharged, affecting quality. Such dissolved hydrocarbons cannot be removed by probe action; their final content must remain below 50ppm at the time of drainage.

Installation

During tank construction, two nozzles equipped with isolation valves and blind flanges were installed above each dewatering line to facilitate in-service probe installation without requiring a shutdown. The manual valves on the dewatering lines were systematically replaced with motor-operated valves (MOVs), one sump

at a time during periods when dewatering was not taking place, thereby minimizing operational disruptions. The complete installation can be seen in [figure-6](#).



Figure 6—Final Installed Tank Dewatering System

Pre-Commissioning, Commissioning & Start-up Activities

The objective of the pre-commissioning phase is to systematically execute all essential steps of the process to ensure that the commissioning phase operates optimally. The ultimate aim is to verify the configuration and DCS programming, as well as to confirm the proper functioning of the integrated unit in accordance with the project objectives.

IR Test after Glanding and Termination

Measuring and continuity testing of instrument and power cables are conducted as described below, with test results recorded and witnessed by a COMPANY Representative. Continuity is verified for each core from end to end using a multimeter. Absence of continuity between cores, between core and earth, and between core and screen is confirmed with a multimeter. Insulation resistance is measured using a Megger meter at 1kV for electrical cables and 500V for instrument cables, between core to core, core to earth, and core to screen; the recorded values must exceed Mega Ohms.

Electrical Tie-In Works and Energization of Distribution Boards

- Verify that all relevant permits and certificates are obtained and valid.

- Comply strictly with the Lock Out Tag Out (LOTO) Procedure.
- Confirm that all sources of energy have been fully isolated prior to commencing work.
- Ensure appropriate earthing systems are properly established.
- Don the required personal protective equipment (PPE) specific to the activity.
- Inspect for loose materials that could contact terminals and cause short circuits.
- Initiate energization and verify that all operational parameters meet specified requirements.

Calibration

To calibrate the probes, two containers, each approximately 800 mm in height with a diameter ranging from 100 mm to 150 mm, are used. These containers are connected to a grounding system via a 12-gauge cable secured with a small welded screw.

The containers are filled separately, one with water and the other with dry crude oil sourced directly from the process. It is essential that the crude oil sample contains minimal water, while the water sample is as free as possible from crude oil contamination. Both samples should originate from the same process in which the probes will be utilized to ensure precise calibration.

Each container should be filled with a volume of sample between 20 and 40 liters. Calibration must be performed at the installation site.

Loop Check

- Loop Check: This phase comprises several distinct procedures.
- Ensure that loop folders are systematically organised during the loop check process.
- For transmitters with electrical connections to DCS or ESD, conduct individual loop checks for each device. This process should also include verification of alarm functionality.
- The loop check encompasses comprehensive testing of all control loops. These tests should approach various loop types from multiple technical perspectives, as detailed below.

Functional Test

- Ensure all the Open & Closed loops are ready for functional test.
- Ensure proper communication while performing the test.
- Ensure all the logic and interlocks are tested as per Operation Control Philosophy and Cause & Effect.

Control Philosophy Functionality Verification

- The Control Philosophy functionality verifications shall be carried out as per the Operation Logic and automatic tank dewatering system control logic given in the [Table-1](#).

Site Acceptance Test (SAT)

The SAT procedure generally begins with a comprehensive visual inspection of the installed system. This inspection aims to verify that the physical installation meets design specifications, safety requirements, and quality standards. Inspectors should check for correct component placement, secure connections, proper labeling, and overall workmanship. Following this, the SAT involves conducting a series of functional and software tests. These tests assess whether the system operates as intended under various scenarios and conditions, ensuring all integrated elements communicate effectively and perform reliably.

Result

Automatic dewatering technology was implemented in a Crude Storage Tank without disrupting operations. Site Acceptance Tests confirmed seamless integration and reliable performance, achieving oil carryover levels below 50 ppm in produced water. The system has operated optimally for over nine months, reducing the need for full-time field operators, lowering OPEX, and mitigating HSE risks. Its success has led to adoption across all existing and new crude oil storage tanks.

The drive to enhance operational efficiency, safety, and environmental compliance in the oil and gas industry has spurred significant investment in automation technologies. Among these advancements, automatic dewatering systems have emerged as an effective solution to longstanding challenges associated with the management of crude storage tanks. Manual tank dewatering is not only time-consuming but also inherently risky, given the exposure to hazardous substances and potential for human error. In contrast, automated systems leverage real-time monitoring, intelligent sensors, and precise control mechanisms to ensure optimal performance and minimal disruption to ongoing operations.

At the core of this successful deployment lies a robust instrumentation framework capable of accurately distinguishing between oil and water phases within the tank. The technology integrates advanced level sensors and programmable logic controllers (PLCs) that enable autonomous decision-making. These components work together to initiate and regulate the dewatering process, ensuring oil carryover levels in the discharged water remain consistently below stringent industry thresholds—specifically, under 50 parts per million (ppm). This achievement not only aligns with regulatory requirements but also signifies a substantial reduction in product loss and environmental risk.

The implementation phase prioritized maintaining continuous operation of the crude tank and related infrastructure. Thorough planning and coordination allowed technicians to install and calibrate the system without necessitating a shutdown or interrupting production schedules. During Site Acceptance Tests (SAT), the automatic dewatering system demonstrated its ability to integrate smoothly into existing control architectures. Key performance indicators such as response times, reliability, and accuracy were closely monitored, all of which met or exceeded expectations.

Over an operational period exceeding nine months, performance metrics validated the system's capabilities. One of the most notable outcomes has been the significant reduction in the requirement for dedicated field operators to oversee manual dewatering tasks. This shift not only contributes to reduced operational expenditure (OPEX) but also reallocates valuable human resources to other critical activities. Furthermore, by minimizing direct contact with hazardous substances and confined spaces, the system effectively mitigates Health, Safety, and Environmental (HSE) risks, fostering a safer working environment.

The positive impacts extend beyond individual site improvements. Due to the proven effectiveness, scalability, and return on investment, the organization has opted to expand deployment of this technology to cover all existing and planned future crude oil storage tanks. Such widespread adoption points toward a broader industry trend favoring automation and digitalization to achieve sustainable operational excellence. It also lays a foundation for further innovation, such as integrating predictive maintenance analytics and remote monitoring capabilities, which could further optimize asset integrity and lifecycle management.

Conclusion

In summary, the application of automatic dewatering technology represents a significant stride forward in crude storage tank management. By seamlessly blending reliability, safety, and cost-effectiveness, it sets a new standard for best practices in the sector. The experience from this project serves as a compelling reference point for continued technological evolution within the energy industry.

Acknowledgement

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1. 2021, American Petroleum Institute, API Standard 650, 13th Ed.